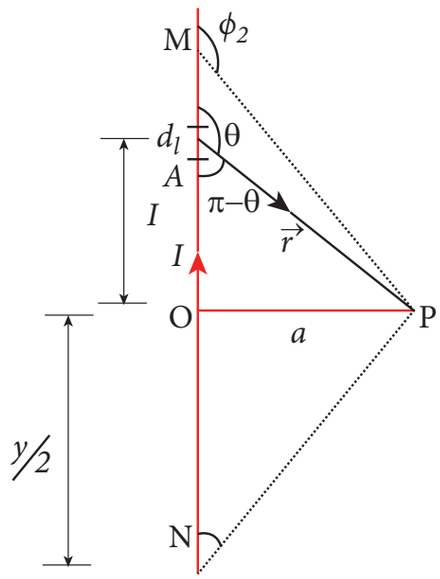
$$\vec{B} = \frac{\mu_\circ I}{2\pi a}\hat{n} \qquad (3.39)$$

Note that here $\hat{n}$ represents the unit vector from the point O to P.

EXAMPLE 3.15

Calculate the magnetic field at a point P which is perpendicular bisector to current carrying straight wire as shown in figure.



Solution

Let the length MN = y and the point P is on its perpendicular bisector. Let O be the point on the conductor as shown in figure.

Therefore, $OM = ON = \frac{y}{2}$, then

$$\cos\varphi_1 = \frac{\frac{y}{2}}{\sqrt{\frac{y^2}{4}+a^2}} = \frac{adjacent\,length}{hypotenuse\,length}$$

$$= \frac{ON}{PN} = -\frac{\frac{y}{2}}{\sqrt{\frac{y^2}{4}+a^2}} = -\frac{y}{\sqrt{y^2+4a^2}}$$

$$\cos\varphi_2 = \frac{adjacent\,length}{hypotenuse\,length} = \frac{OM}{PM}$$

$$= -\frac{\frac{y}{2}}{\sqrt{\frac{y^2}{4}+a^2}} = -\frac{y}{\sqrt{y^2+4a^2}}$$

Hence,

$$\vec{B} = \frac{\mu_\circ I}{4\pi a}\frac{2y}{\sqrt{y^2+4a^2}}\hat{n}$$

For long straight wire, $y \to \infty$,

$$\vec{B} = \frac{\mu_\circ I}{2\pi a}\hat{n}$$

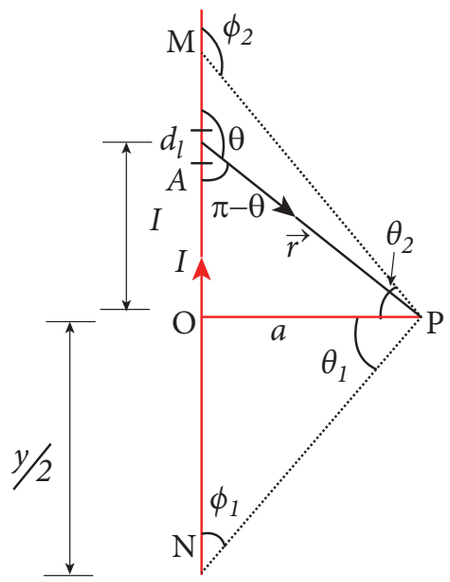
The result obtained is same as we obtained in equation (3.39).

EXAMPLE 3.16

Show that for a straight conductor, the magnetic field

$$\vec{B} = \frac{\mu_\circ I}{4\pi a}(\cos\varphi_1 - \cos\varphi_2)\hat{n}$$

$$= \frac{\mu_\circ I}{4\pi a}(\sin\theta_1 + \sin\theta_2)\hat{n}$$



Solution:

In a right angle triangle OPN, let the angle $\angle OPN = \theta_1$ which implies, $\varphi_1 = \frac{\pi}{2} - \theta_1$

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